

AI Workbench

Cost Optimisation Playbook

From \$\$\$ GenAI → Cost-Intelligent AI Systems

1. The Problem

AI cost is not just model cost

Total Cost =

- Compute
- Tokens
- Storage
- Orchestration
- Human effort

2. The Strategy

Shift from:

 Always-on + model-heavy



 On-demand + retrieval-optimised + governed

3. Core Principle

“Move work out of runtime wherever possible”

- Pre-compute > real-time compute
- Retrieve > generate
- Cache > recompute

4. Architecture View

laC → Local Runtime → RAG → Control Plane → Apps

↓

Cost Optimisation Layer (every step)

5. Key Levers (Executive Summary)

1. Zero idle infrastructure (laC)
2. Retrieval efficiency (biggest ROI)
3. Caching (near-zero marginal cost)
4. Model routing (right-size intelligence)
5. Pre-computation (shift left)

6. Outcome

Traditional: High cost per query

Optimised: Low cost per query → near-zero at scale

--- TECH SECTION ---

7. Infrastructure Optimisation

IaC → deploy on demand → destroy after use

- No idle clusters
- Event-driven execution

8. Local Runtime > Managed Services

- Local LLM (LM Studio / Ollama)
- DuckDB + FAISS

9. Model Routing

Simple → small model

SQL → medium model

Narrative → strong model

10. Retrieval = Cost Engine

More chunks → more tokens → more cost

11. Semantic Chunking

Fixed chunks ❌

Meaning-based chunks ✅

12. Hybrid Search

Vector + Keyword → precision + meaning

13. Top-K Optimisation

Before: K=10 → high tokens

After: K=3 → low tokens

14. Caching Layer

Q→A, Q→retrieval, embedding cache

15. Pre-Computation

Compute once → reuse forever

16. RAG > Fine-Tuning

Lower cost, better governance

17. Vector Storage Strategy

S3 → FAISS (on demand)

18. When to Use OpenSearch

- high concurrency
- large datasets
- filtering

19. Tiered Vector Architecture

S3 → FAISS → OpenSearch

20. Evaluation Cost Control

Sample, target, reuse

21. Governance = Cost Control

Policy before LLM

22. Storage Lifecycle

Prune, version, compress

23. Event-Driven Execution

No query → no compute

24. Human-in-the-loop

Targeted escalation only

25. Advanced Pattern

Cost-aware query planner

26. Final Architecture

Infra → IaC

Compute → Local + dynamic routing

Data → RAG

Retrieval → Optimised

Runtime → Cache

Storage → S3

Control → Workbench

Evaluation → Targeted

27. Key Takeaways

- Retrieval drives savings
- Cache eliminates marginal cost
- Avoid always-on infra
- Shift work left
- Be cost-aware

28. Closing

Cost-intelligent AI systems